

ActInf Textbook Group ~ Cohort 4 ~ Meeting 8

(Chapter 3 part 2)

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Duration: 55:51 | Uploaded: Unknown Transcript method: auto_caption

okay it's cohort four and July 25th and we're the second discussion of chapter three so before we look at the questions is there anything that like anyone just wants to start anything about the reading or anything even maybe some recent uh live stream or or course or anything or just what their reading is like now of chapter three yeah go for it so I I was thinking about this idea of synchronization between the inner model and the outer model which another term for that could be adaptation to the environment active adaptation modifying the environment and building a model of the environment but Bronwyn mentioned personality disorders and one of the uh features there these are very fuzzy Constructors being stuck in maladaptive patterns that don't really fit the environment and person is unable to change them thus their energies uh they swim in free energy so to speak and they suffer and there's negative effect that is present all the time I I do think that these models are a little too straight and narrow to account for Psychopathology and I do think trying to explain something like that with sort of a a need formula is is just not ready yet I mean I I think that these are not doing it justice because you know uh people exist who spend decades in maladaptive behavior and they don't uh do much about it to get better so I I just find it that you know when when we dictated and say this is the law this is the first principle and this is how the living thing does it bacteria perhaps but in a person with like malignant narcissism no they don't yeah it's a really good uh question like what is the role of the math in the science what's the role of the science quote science what's the role of the math in the science how just broadly extrapolative is it in its interpretation um and then what's the role of just any any how is how is a math and science mobilized with personality what does a principle matter in the specific what does a specific matter for a principal those are all huge questions I think you'd be interested in a cybernetic Big Five Theory they have a few papers where they connected to free energy principle to try to go at different psychopathologies and try to see how you can like think of in a cybernetic perspective it might be something you might be interested in I'm familiar with the big five

Theory it's also statistics based I'm a clinical psychologist and I think it's uh has nothing to do with personality disorders I think it's again trying to uh uh quantize but but still you know we're going for mathematics trying to Define I mean it would be good to to bring somebody in the room who has a borderline personality disorder and try to apply that theory onto them they're predictably unpredictable there's something missing like I don't know field Theory or Chaos Theory framework it's just it it it uh it doesn't match my empirical data of interacting with his patients on a daily basis you know and it doesn't come from papers it comes from seeing what they do in life over and over and over again it doesn't work they still do it so uh yeah one way I I I might kind of ask this would be the personality traits are more about the um characteristics of the paths in a cognitive model so it's like certain type of Dynamics in the map not the territory then in the you know maybe a clinical setting first off it's engagement directly rather than engagement with the model and also there's perhaps less of a focus on just like um in the in the short term about just um population scale patterns and maybe more emphasis on a person's action patterns and so then how can these two sides kind of support each other in an applied cognitive science rather than like disengaging [Music] characteristics and uh cbtf cbft uh and cybernetic Big Five Theory as like another layered uh outside of just traits what was the name of the thing outside of traits uh characteristic adaptations oh yeah also anyone of course can always be just clicking on the cursor and following and taking notes like people can have things just it's all good if people want to to discuss or just filling in other angles but thanks everybody who has shared any other thoughts before we look at any of the questions there could be anything um actually yeah it just as a brief follow-up to the preceding conversation I was just curious if anyone finds interest in the notion of like hyper precise priors I see that come up very frequently in uh applications of active inference to Psychopathology and um I think basically the idea that there's a like a very strong belief that a certain model is correct to where it kind of skews and coming um you know sensory information and it kind of continues um to to impact future actions um despite like disconfirmatory sensory evidence I mean I I'm just in response to the earlier conversation I found I've found that to be really interesting and I thought it dovetailed really nicely with with studying Psychopathology actually yeah I I think it matches well to something like delusion because if I believe firmly that neighbor walks in my house and changes the furniture when I am outside then and it doesn't get uh updated based on me calling the police and calling you know installing cameras then that sort of hyper precise priors exactly but I I do think that again trying to stretch these approach to different kinds of Psychopathology like as I've

mentioned say personality disorders right so a lot of personality disorders are uh related to early experiences where the child or adolescent try to adapt to the environment which is very difficult to survive it emotionally and they've developed some habits that were not solutions back then but there's nothing better that was available to them and they carried it through into adulthood so uh for example you know a child often in a divorce of parents ends up thinking that he is the reason why they're divorcing they end up internalizing distress and saying I'm the guilty party and then they get in trouble because if I'm in trouble then maybe these two Knuckleheads will get together and then you see an adult patient Psychotherapy in their 30s that apologize every session for everything you know and that's an example kind of and it's not necessarily I mean it has a completely different flavor from delusion and I don't think the language is hyper precise priority is sufficient to distinguish between the two there's qualitative differences uh and so I guess that more stuff is needed to describe the difference of it what about something like uh obsessed impulsive disorder uh where you could think of it as a from like an ecological psychological perspective as uh a low tolerance for affordances so yeah low tolerance for uncertainty basically that that would not do it justice that would be a very simplistic approach to OCD because there's different subtypes people who do washing and check-in are very different from borders so uh uh you could have people dysregulated and people were just regulated childhood OC there's a lot of it all together this is an example of trying to you know uh fit something very heterogeneous complex into a simple formula and just saying oh we got this cool Theory let's just explain OCD with it no I mean I think if we want to do the OCD Justice we need to start with a lit review go to PubMed and read all the theories about OCD genetic influence you know environmental influence uh you know chemistry everything else put it all together and then we should do it so this approach from first principles to something as complex as uh it's almost Farr is a psychiatrist uh Colonel friston is a psychiatrist a lot of people who aren't they want to just you know schlep a formula to OCD and it just it doesn't work well again what what is really the role of a of a specific model uh construction of those models is usually not uh utilized as a full account but maybe in it it can seem that way or be used that way in certain settings like to guide action and then also like one good thing about generative modeling is that it allows really specific discussion of what has and hasn't been specifically modeled so first principle is just because something feels like consilience in one's own mind does not equate to an explanation let alone useful in the world then there can be a discussion of like okay what um sensory modalities and interoceptive modalities are taking into account and what how's childhood you know in this specific model

of these people who have ocds like oh there's no generative model well we have chapter six we have chapter nine you can build the model but definitely if it's just speculation about what inclusion or exclusion of variables you're just like kind of casually curious about or professionally curious about it's all speculative so I think that kind of supports a more bright line because then people can just spitball a little more broadly structurally which everyone's gonna have some expertise in some area but then also it's clear like when someone's not referencing a specific generative model it's just a pure hypothetical so how would uh what what um what would a hypothetical linear regression account for nothing so what would a hypothetical cognitive model account for well maybe something but doesn't have to so I have a question is this more of an issue with how we're classifying psychopathologies or is this as in like each classification is too broad perhaps and we need to be more specific maybe and because maybe like the same like you were saying like there's many different types of obsessive-compulsive disorder so maybe the ones that pertain to uh a low tolerance for uncertainty would need its own specific classification versus other ones that are different yeah which is a question there there's many issues there one is you know we apply mathematical apparatus to another field we're trying to either or cybernetic theories or something else and it doesn't always work for example in Psychology there's something called ambivalence which is hating and loving the person at the same time that concept violates the logic of Aristotle and all of mathematics is based on Aristotle and logic so there you go you can't use the absolutely fundamental framework while you're trying to use something more sophisticated like classifications and whatnot and there's many critiques of DSM classifications as the declassifications but the important thing is the construct is complex and then you know before trying to model it or explain it we should study the existing literature and and go from there because otherwise we are trying to model our fantasy of OCD and what sort of the the Practical reality of OCD that is you know like uh sadly it's it's uh still very bad like schizophrenia we know about 14 possible causal factors but there is no coherent fear of schizophrenia that everybody exists on nor depression it's the oldest condition in the history of medicine but there is no integrated theory of depression still it doesn't exist and so this active inference is one of the attempts to do that but it still falls short like if you use active inference and you try to explain everything from autism to I don't know motor disorders it doesn't fit it doesn't capture the the core of autism it just it it uh it seems to fit you have this resonance you're like oh that's cool I think I've explained it but if you actually see people with Autism you see that you're explaining something else I mean because the core of autism is

mentalization deficits so uh I I don't know if it's a good answer but it's a it's a very messy field it's very different mathematics is very kind of you either proven theorem it's either true or false in political psychics and there are different things that's why uh modeling work is part of an integrated scientifically plural framework I need to do both oh just just it's it's a suit it's a super over fit like it's always about reading a monks in between the citations and everything else it's not about one paper one experiment one data set it's always about being explicit about the uh interpretation and how the researchers model was made and so if somebody shows up talking to you um with a lot of personal experience and or a lot of meta-analysis I'm sure that's a productive conversation also can be productive um someone who might not be aware of some of those complexities but yeah that's also there's also just like these are it's always like active inference or is this about just like any kind of modeling numerical or statistical modeling because someone could also make I I believe in identical we're not linear regressions our health records aren't linear regressions that's not the body that's not decision making it's just linear regression so Decades of insurance being calculated that way for example so if that ends up using cognitive modeling that that's kind of the gray Zone slash question is what is that kind of cognitive digital twin not just the accumulation of information from a privacy perspective but that kind of like more decision making representation which of course is still just map and not territory but that's a different area just one kind of modern connection I don't know do people have any thoughts on that or how this relates to um different machine learning like how does it relate to machine learning or what's happening right now with with synthetic intelligence um well I know that uh but it seems like self-attention mechanisms are also trying to minimize perplexity and there was this recent paper that they're using called like evolutionary Transformer architectures where uh they basically add like an evolutionary block to it which allows it to so you have multiple sort of models and they're also all trying to minimize perplexity and then I know like GPT the latest with GPT was they it was like leaked that they're using they're using a it's called a mixture of experts or yeah basically like a similar type of algorithms to just find what comes up with the best sort of output among many different models so I just thought that's kind of similar to like some things that were mentioned in active inference especially this chapter yeah that's one great connection also with Collective behavior and evolution and active but also attention I know Holly and uh others have have kind of thought about this or or considered a little bit like attention mechanisms first off attention is just one of a huge variety of cognitive mechanisms anticipation expectation attention um this is we can think about the regimes of attention which is a concept from the

cultural work with in cultured patterns regimes of attention uh to sensory into different things and there could be a Transformer with a real cognitive model explicit semantically interpretable active figure 4.3 type cognitive model here's the explicit attention function for example and that and that there's a lot more say in that area but that's for those who are interested I feel like that's a cool you know thing to look into any other chapter three thought or we can go to the questions all right everyone can can vote or just also open the question all right note that the resulting preferred course of action or policy is directly measurable in experimental settings whereas a value function or prior belief needs to be inferred and hence is more indirect if not tautological measure I'm not sure how or why the measure might be tautological in the case where the value function needs to be inferred what is the tautology here anyone have a thought on that all right well I I I am not exactly sure but we can contrast reward learning with active inference type generative modeling so reward learning you're going to assign some utility or value to different states and then pursue rewarding States like different chess pieces get a value and then you want to maximize that expected value that's the imperative for selecting different actions then inactive action is going to be selected based upon it being the most uh basically it depends on the kind of agent because different agents have different kind of relationship with the future they're simple agents or just um kind of inert or have deterministic action policy but then there's the whole open-endedness of cognitive systems so they act to learn but they also act to realize their preferred uh their preferences about sensation but however you look at their behavior whatever you think about their behavior whatever you propose as as guiding their behavior it is what is directly measurable in experimental settings which is shown in figure 9.2 oh 9.1 which we talked about actually in the um other cohort earlier today so we're talking about what we present as stimuli and what we observe as Behavior so that's what this modeling approach is Tethered upon that setting where it's being a cognitive model is being projected in contrast if you propose some second order function about the behavior like well it likes being in the light not because it expects to be the light and that's the most likely thing but because light is rewarding and so then it's going to pursue reward you've basically proposed an ad hoc potentially principled but at the very least indirect and possibly just self describing something that isn't this though maybe there's some specific case but it seems pretty vague I mean anyone else think about the tautology or I sort of uh just took that that quote as referring to like uh so preferences pardon me I'm still to kind of trying to wrap my head around this like like you're already starting with like a set of preferences I suppose that you're c um Matrix right and um so there's the tautological aspect is

that I suppose whenever you're modeling you always have to start with some kind of C like we it's it's as if it's saying that active inference Is Not An Origin story it's just like we're already there and so like there's no explanation for the starting point we can only measure what maybe results following after our priors are already defined something along those lines nice that's a great thought what there's a few ways to look at it one one on the more formal side is like if we think about things that don't plan they're going to be amongst other things like that just the same and all like diffusing particles so like they have no preferred state in the gas box um so they're they're like they're behaving like just kind of like classically from experimental perspective or like according to thermodynamic equilibrium whereas systems that have an attracting set homeostasis or um acting with the c to kind of anchor the attracting set without any pragmatic value policy selection there's just policies are driven by pure epistemic which in abstracted settings maybe on optimal situation but for embodied settings that's not the general principle so that's why there's those two pragmatic and epistemic which helps give a lot of degrees of freedom because this is a value that's calculated for the specific model that somebody constructs like in chapter six so it can't be just generically said like well the free energy of an estimate goes down when it gets a seed it's like it's it's way way speculative to say before having like the specific kind of model that's even being discussed so and it's just it's explicitly what is being um the attracting set itself not a proposed second order variable just one possibility but under another approach is like that is like backs like is it about modeling or is it about active inferences like okay what other what other approach can be taken in this setting is this not a worthy setting or like is there another way to approach this and what would that what would that kind of a variable or what would that model be so it doesn't have to be all only every model but that's kind of the the golden Hammer Golden Nail or whatever now that can be a simplifying cognitive heuristic which has all the benefits and trade-offs of any cognitive system engaging in a similar heuristic but we're in the textbook group so we can like learn and like scrutinize it or like use it in ways that that don't necessarily make specific sense in any given one person's experience of using it leaf blower any other thoughts on this question what are the minimal contents of the model encoded within the internal states of an activation I'm presuming that one a statistical model of the hidden causes of a sensory data and two preferences about the state the generative model needs to be in order to ontologically persist Through Time I feel like one is specified in the section but where in the generative model are the preferred States located who has any thought on this so would this not be what they describe as like the motion of the action States as the prediction of the preferred States as

well I don't know if that's that covers that aspect okay possibly like in the actual person I think it kind of can you say the relationship again yeah so in the book they describe uh how the Markov blanket not only is predicting uh sensory straight States it's also doing so for the action States as well I don't know if I'm phrasing that correctly if I can find it one equation based way to to look at this is so where in the generative model are preferred States located there the sea here so they're used in the free expected free energy calculation so if if we're talking about a kind of of simple model not a simple system but a simple model where the cognitive states are just like a pass-through or you know a divide by two or you know there or it just is just a noise so the action policy in the sensor are uncorrelated or uncaused or anything like that like those are all special cases but action policy selectors which might be one step selectors like instantaneous selectors doesn't have to be an explicit planner per se um could be like a gradient based system with bacteria um pragmatic value is where preferences are loaded and pragmatic value rebates policies accordingly to how close you get to the preference Vector slash variable and how relevant that is in the behavior is situational there's no generic answer because the other component is about epistemic gain and so this is the the relationships that's described in all those special cases where this or that doesn't apply so it's a totally good question but it's a very you know this is one of the key differences between reward learning and active inference in active inference the preferences are explicit over sensory States that's a relationship with perceptual control theory and it prevents overfitting because it's about the incoming sensory data predictive processing all of this um there's no proposition of a reward or utility function what the agent is supposed to be doing what our inference does procedurally is just these optimizations on the specific proposed generative models so first principle may do last work generically principle of least action people are not always bringing this out to talk about well this is why this plane did or didn't fly well this one was just following its path of least action so that's not an explanation or an account and I think that that's that code in what you shared Lexi like sort of about like about uh particulars and generics and everything like that first principle less specific work done by the framework understanding the framework very prerequisite for modeling understanding claims for for of course just the joy and the fun everything like that all the fun we have um but understanding claims like evaluating whether a given um usage of the active inference ontology is like valid but then to do the applied modeling and to understand it in the way like that a graduate student type might then these are only kind of the enabling pieces kind of like the L2 least um some squares minimization has a relationship with linear modeling but a more statistical student

might learn more about the underlying basis of the linear regression more empirical statistics student might just apply data set modeling not go too deep into the theory just want to kind of provide some of that context because like Kristen at all have explicitly said this is for this textbook that that were gathering around like that it's a kind of Masters student again taped out for what it means and there's always more variability like within them between all these other things but Master students getting on the way to doing modeling in this approach based upon reading the epistemic first half of the book I don't know what piece of course we're having fun figuring it out and then going into the modeling in parallel following along after the chapter six with one's own kind of specifications or just like Explorations for different systems some stubs of which we have and we can collaboratively do too but it's kind of just a general comment because like if the philosophy is is amazing it really is and people bring so much to these discussions that that I'm just so glad that that we have them and everything like that and a lot of it is is very uh General and importance outside of just active inference which is really like uh in in the in the textbook sense like uh comparatively can be narrow because it can feel like there's like a lot of like things to get like rabbit holes to get lost in and everything Craggs to fall into I I hope it throws a little context hence just like regularly working with material and just broader sketches natural language that is consistent with the equations but you know there's only so much mystery in one's time in in life with a mystery with the equations Alexi yeah um thank you so much I mean I think you described it beautifully I just wanted to also comment that everything we're talking about only applies for systems just like when it says early in the chapter that have an attracting set this is fundamental quality of the systems you describe on this figure 3-3 when they show a system that doesn't have an attraction the system that does have an attractor active versus inference models talk about the ones that do have an attraction okay kind of converge right uh and so I recently asked Carl via email and he asked lanceland Acosta as well what if they're attracting set changes for example if I sleep there's one attraction set when I wake up there's another tractor set and my entire system bifurcates it evolves from let's say periodic attractor to a strange attraction and they are working on that right now I think this is a work in progress but they will create a sort of a hierarchical structure where the attracting set itself is not kind of set where you can describe and say that's the attraction set but it can actually change in and of itself and that's that's a new interesting development that may uh that may help I don't know how people feel about that yeah that's awesome that sounds really cool one one piece to bring in here this is Axel Constance like extremely um kind of uh girdle Escher Bach type uh description of fep which kind of just had

to be said but he laid it down and the simple point I'm making this paper is that the free energy principle is not concerned with the sufficient conditions of existence but rather with what must have been the case given that you exist it's not about figuring out what it takes to be alive it's about figuring out what took you there that's just so deep and I think it gives a difference accounts for a lot of settings just like the surprisal versus reward does that um now that that in place I don't know exactly how they're related here because maybe this could be taken you know but it's like saying we're making a measurement theory for measurable objects we have the blur so we know when there's no signal that's just noise that's going to be because that's the Baseline informationally and then we're going to have a theory of measurements an analytic framework for measurements cognitive measurements observables and unobservables a four measurable systems or if you do a hypothetical it doesn't have to be like an Adaptive system so that's what's fun in the textbook group is like we can imagine just like okay let's just imagine that the external state is just alternating between two states it's just like I mean it's great to really wonder about these um really contemporary examples and everything like that but like pedagogically a lot of these existentials are non-relevant it's good to keep them in mind because I think a lot of the the technical part like people can study and um and the math learning group and hopefully like it's you know there could be good math discussions but this is like I'll bring so much of the broader context in what everyone shares but because the the math is not a metaphysic and this inversion of of uh conditioning like most people if they think about um a theory of everything they're going to think about it like from a sort of material up like how things came to be versus um so versus at least the measuring or modeling framework for Signal versus noise which does a less slash different work per se than the than the um the accounts of how something like comes to exist because different mechanisms for things coming to exist they might have from it for a given generative model like they might have similar or different Dynamics but the model is not gonna like I think here to um JF classier at The Institute in his work with the uh symbolic active inference robotics and about how we can have different implementation architectures that can be modeled from a behavior like from an active inference Behavior Uh perspective but this doesn't mean that everything has to be like running this exact statistical model as a kernel that's like figure 4.3 is like super schematic extremely it's like it's it's not a claim about a system it's just super evocative about the kind of structure that can be expressed for in the most minimal way and then there's a lot there's a lot more to do and everything but that's why it gets kept it keeps on getting returned to because it helps give a first pass Separation on like priors sends to Hidden as hidden state to

sensing World model and action model so these are some of the essential features of of as it turns out a nerd and sentient creatures but this is not like I hope that's not too too contextual but just want to clarify because I I can I can imagine slash C that that what the equations do is a primary concern slash awareness and uh so so I think that it's very um worthwhile for people who want to continue like learning and thinking about it like um sharing their perspectives on what these maths actually do because then it's just parallel streams where one is just like like now I don't know just what do people think about that or or how do we really connect up even just in these short discussions some of these mathematical details which are kind of just like just defined as true essentially and all these other systems of Interest since like ways of knowing that we all are interested in I hope I'm not too like far afield from the way that things are being framed a moment ago but um I mean as far as something like uh simultaneously or simultaneous like attractor sets that might be competing with one another for example um and the idea of like you know you have a series of preferences um I've I've not had too much trouble wrapping my mind around the idea that you can have you know competing uh tractor States or competing preferences if you need to use the restroom and you're hungry and you're very sleepy all at the same time and so we have these models that you know whether it be the discrete version or the continuous version they all happen in time and they all involve like policy selection over time so for example if I decide oh like you know it's I'm very tired but it's only 2 p.m I could take a short nap and then I could get up and you know have a snack and use the restroom whatever I need to do but you can plan things out over time and and as far as the relationship between that and the math goes um like we have that nice like expected free energy equals you know epistemic value the real you know the combination of the epistemic value and the pragmatic value and so your C's your preferences are only in um pragmatic value as you mentioned earlier Daniel so whenever you aren't sure what to do you have these competing preferences like there's a almost like a uniform probability between them like which one is going to win out I'm not sure you would rely upon doing actions that involve gaining more epistemic value like oh well I'm going to have to spend a moment like planning and what makes the most sense here can I come up with a plan that helps me to like resolve some of this uncertainty going forward and do that and you know um or maybe you have a certain belief that one of these things is just incredibly important and so you you stick with that one in the first place or maybe you become conflicted not to not to backtrack but on the topic of Psychopathology I'd be interested to see someone just attempting to model decision making and planning and so on in the context of someone who might have you know some

kind of anxiety disorder for example or in fact like maybe you have an additional Merchant behavior that is spending you know five to ten minutes or maybe a few hours like ruminating having issues about what to decide upon I read up I'll try to find it but I find an interesting paper that tries to get at you know modeling uh rumination in that kind of context so that those are all my comments here on the idea of like competing preferences and relationship to the mathematics oh nice thank you one other um aspect uh to kind of so whether it's the visual form which is describing the sparsity of the base graph like which nodes connect to wish knows this is this is kind of like the skeleton of the model generative model this is where the time concept is embedded this is where the action concept is embedded sense making concept prior concept so it could be viewed this way and then the equations are describing like statistical operations on that and and that's why uh whatever structures proposed it to be applicable to any actual system it's a parameterized data set specific model specific data set like saying let's um imagine that uh childhood just imagine the childhood experience one two and three are associated with um with cybernetic B5 or DSM such such and such just imagine that they're that case so what do you think about this being the case this one's associated with that it's like well I don't know you haven't specified the model or applied it to a data set but it's like oh yeah but you can like see that that would be a really important thing right it's like okay perhaps but that's a purely verbal argument that's what I was referring to earlier as just speculative that can be an intuition pump and I believe serve a lot of other functions and and communication functions and other features but um that is not what a scientific account consists of which is a much slower term um pluralistic um synthesis not not to be like too old school on that topic but I think it helps enable open-ended discourse which I believe is important and when people watch so many different backgrounds and and and familiarities that that of course everyone's going to be like thinking and saying things that we would not think and say because we come from so many backgrounds but just what happens in a discussion around a textbook is not um is not it's in a Continuum of communication but it's not of the type of a scientific account ajith yeah so um I have a question about something else but uh just kind of the idea that like our behavior is governed by a principle of least action so how does this affect like um assumptions of like measurement Independence so like some experiments uh for example like the one they use to sort of even tests so I'm not my background isn't physics so um if anyone knows correct me if I'm wrong but to test like uh Bell's inequality there's an assumption of measurement Independence that like you can actually make decisions and it's not actually affected by sort of the environment or what's

going on uh and this is sort of a way to actually like make it so you can actually test certain things and if you can assume measurement Independence that kind of breaks science there or a lot of people think it breaks sort of science there so uh Concepts like super determinism uh is one that assumes that every sort of like Behavior or action is something that uh kind of is dependent on sort of all the past States as well basically so um yeah I was just wondering like I I don't know exactly how like active inference would play in that like does this go against measurement Independence if it is a principle of least action because it's like going in the direction like the most least effort basically yeah there's a ton of areas I'm not familiar with every single aspect there this is as Toby Smith um uh Sinclair Smith uh called it in his recent dissertation which was uh live stream 54 series he called it compositional cognitive cartography we're talking about maps variational free energy expected for energies are properties of maps that are human constructed and modeled hashtag John Dewey all of this other pragmatism and and threads that are brought in there's nothing generic to say about free energy or expected free energy except for the truly generic active ontology type Expressions so that's why we can like open up and have a lot of richness with the chapter six type discussion about modeling because they're so far to go as people are encouraged to do in the textbook group um and through time with just like whether you're coding is that um this or that I hope we can develop better uh coding and all these other things collaboratively anyone's welcome to do a lot of these are questions about specific model settings and um there's a big Continuum between kind of touching the theory and touching the system but that's the inclusion that we get to cultivate with the right Niche if we can create the right artifacts that support people who are different understandings because like there's a lot of threads that were brought in relatively early in terms of like the connection to ecological psychology and predictive coding some of these things are are only finding any empirical uh um support but with rapidly expanding amounts since like 1999 with kind of the classic round Ballard type predictive coding models so not there any given communities like acceptance of something matters or what is in the literature because there's kind of like the timelessness of the ideas in a sense but like again I just there's a broad range between the mathematical and the engineering and all these other perspectives so it's very wide that's what makes it wide open in terms of like creating like well what about accounting from this perspective that's those are almost always good directions to go if you're you know here and curious and uh what What's so fun is as people continue to go deeper in their um work that it it can go from like kind of uh tautologies um or or just just um just active infantology scaffolded to actually implement it

which brings us a lot of um and and entails many codes of paint and a lot of like Hands-On and working with some of these questions that are that we may have surely the agent is a generous model map territory but then people sometimes do go beyond and go into the ontological and and uh but but that's that's definitely a secondary Claim about a system one that might be merited or not but it's a secondary claim so first approximation we're just talking about the modeling apparatus that may describe something fundamental in a given system but also literally may not or just maybe one of many things that any other um thoughts or questions people have yeah well a few more things from chapter three and then into chapter four anyone who has read it before Wanna Give a note on chapter four M Carl six any other questions or or we'll end all right thank you see you all next time bye